Zafar Ahmad

Cardiovascular signal processing · Electrical engineering

ahmadzafar577@gmail.com · Pakistan

LinkedIn: zafar-ahmad-ab87b6183 · GitHub: ZAFAR-AHMAD-5359

Google Scholar: D5W9TVwAAAAJ · ESC 365 profile: esc365.escardio.org/person/1388860

Research portfolio: zafar-ahmad-5359.github.io/research.html

RESEARCH PROFILE

Electrical engineer (MS, NUST) studying cardiac electrical activity and its acoustic and pulse consequences through electrocardiograms (ECG), phonocardiograms or heart sounds (PCG), and photoplethysmograms or optical pulses (PPG). My work combines clinical acquisition, signal modelling and wearable research. I supervise CAREON apprentices and collaborate remotely with Qatar University.

I seek doctoral training in physiological modelling and prospective validation: can signal-derived cardiac timing constrain patient-specific physiological models, and how do measurement uncertainty and recording-device differences affect identifiability?

EDUCATION

MS Electrical Engineering — NUST, SEECS

Islamabad, Pakistan | September 2023 – May 2026

Major: Control Systems. CGPA: 3.20/4.00. Thesis grade: A. Degree conferred 18 May 2026.

Thesis: Cross-Modal Reconstruction of Cardiac Signals: Decoding ECG-PCG Electromechanical Coupling Dynamics.

Supervisor: Nazia Perwaiz; **co-supervisor:** Muhammad Salman Khan.

BS Electrical Engineering (Communication) — UET Peshawar

Peshawar, Pakistan | September 2018 – November 2022

CGPA: 3.21/4.00. Final-year project grade: A. Degree conferred 3 November 2022.

Thesis: Automatic Detection of Paediatric Congenital Heart Diseases from Phonocardiogram Signals.

Supervisors: Syed Waqar Shah; Muhammad Salman Khan.

RESEARCH INTERESTS

Cardiac electromechanical coupling; valvular phenotyping and phase-aware murmur analysis; interpretable cardiovascular biomarkers; multimodal wearable sensing; signal-informed physiological modelling; generalisation across sensors and populations.

TECHNICAL SKILLS

Programming	Python, MATLAB; PyTorch, TensorFlow/Keras; SciPy, librosa, PyWavelets.
Signal analysis	STFT/CWT, spectral density, MFCCs, envelopes, peak detection, resampling and artefact handling.
Modelling	Conditional GANs, attention models, CNNs; copulas and coupled hidden Markov models (ongoing research).
Evaluation	Cross-validation, ablations, calibration, uncertainty, subgroup and cross-device analysis.
Research tools	Git/GitHub, Linux, GPU training, LaTeX; ECG/PPG/IMU sensor integration.

RESEARCH EXPERIENCE

Research Assistant and Team Supervisor

CAREON · NUST, SEECS / SMV-Lab SINES | December 2025 – present

Continuous AI-Remote E-Health Operations Network. Principal investigator: Nazia Perwaiz.

- Lead technical work on a wearable prototype that records time-aligned ECG and PPG with inertial sensing for motion-artefact analysis.
- Supervise three research apprentices across embedded firmware and sensor integration, signal processing and the cloud data pipeline.
- Set the sampling/acquisition plan and specified an external analogue-to-digital converter (ADC) fallback for precision constraints; develop ECG–PPG timing features and coordinate consenting-volunteer recordings.
- Led a blind cross-device audit of the PCG-to-ECG reconstruction model on 338 independent-device record pairs for the JBHI revision; it did not establish successful transfer to that hardware.

Research Assistant (Remote)

Qatar University, College of Engineering | May 2023 – present

Multimodal Cardiac Signal Modelling. Supervisor: Muhammad Salman Khan.

- Developed band-selective augmentation and MFCC-based 1D-CNN classification for cardiac recordings, published at EUSIPCO 2024 as first author.
- Develop copula-based coupled hidden Markov models linking ECG waves and heart-sound phases; investigate their dependence across recording conditions.
- Analyse synchronous ECG and PCG using spectral, correlation and rank-based methods; explore noise augmentation that preserves physiological timing.

Research Apprentice

NCAI, AI in Healthcare Lab, UET Peshawar | June 2021 – May 2022

Supervisor: Muhammad Salman Khan.

- Worked with cardiologists on paediatric heart-sound recording at Rehman Medical Institute, Lady Reading Hospital and six schools, with echocardiography-confirmed annotation.
- Developed collection procedures, consent documents and clinical questionnaires for the institutional review workflow; participated in school screening and referral.
- Built preprocessing, pitch-shift augmentation and 1D-CNN models for paediatric congenital heart-disease detection, published as the first-author ICAI 2023 paper.

CLINICAL AND EMBEDDED RESEARCH

Electronic-stethoscope field recording; clinical collaboration and annotated cardiac datasets; collection protocols and consent workflows; ESP32, AD8232 ECG, MAX30102 PPG and MPU6050 IMU integration. Planned prototype tests cover timing fidelity, acquisition limits and sensitivity to recording hardware.

RECOGNITION AND LANGUAGES

Best Summer Intern Award — NCAI AI in Healthcare Lab, UET Peshawar, 2022.

English (IELTS Academic overall 6.5, January 2023); Pashto and Urdu (native).

PUBLISHED PEER-REVIEWED CONFERENCE PAPERS

Zafar Ahmad, Muhammad Salman Khan, Muhammad E. H. Chowdhury, Susu Zughaier, and Wanis Hamad Ibrahim. "Phonocardiogram Classification Based on 1D CNN with Pitch-Shifting and Signal Uniformity Techniques." *32nd European Signal Processing Conference (EUSIPCO)*, Lyon, France, 2024, pp. 1536–1540. doi: 10.23919/EUSIPCO63174.2024.10715037.

Zafar Ahmad, Muhammad Zeeshan, Arshad Sohail, Fazal-e-Haq, Muhammad Haris, Misha Urooj Khan, Sayed Shahid Hussain, and Muhammad Salman Khan. "Automatic Detection of Paediatric Congenital Heart Diseases from Phonocardiogram Signals." *3rd International Conference on Artificial Intelligence (ICAI)*, Islamabad, Pakistan, 2023, pp. 188–195. doi: 10.1109/ICAI58407.2023.10136668.

JOURNAL MANUSCRIPTS UNDER REVISION

Minor revision · IEEE Journal of Biomedical and Health Informatics

Zafar Ahmad, Nazia Perwaiz, Muhammad Salman Khan, Sidra Sultana, Fatima Abdullah, Hamza Anjum, and Izharul Haq. "PCG-to-ECG Signal Reconstruction Using Conditional GAN with Multi-Scale Attention." Submitted March 2026; revised manuscript resubmitted July 2026. Under review; not accepted or published.

Major revision · European Heart Journal – Digital Health

Zafar Ahmad, Nazia Perwaiz, Muhammad Salman Khan, Suleiman Alabdallah, Sidra Sultana, and Izhar ul Haq. "Synthetic-ECG-gated phonocardiography for paediatric heart-murmur characterisation: a cross-modal deep-learning method." Submitted August 2026; major revision requested September 2026. Not accepted or published.

ADDITIONAL MANUSCRIPT UNDER REVIEW

Muhammad Salman Khan, **Zafar Ahmad**, and Hinna Nayab. "Parallel multi-scale analysis for paediatric heart disease classification from phonocardiograms." *IEEE Journal of Selected Topics in Signal Processing*. Submitted November 2025; last reported as under review (status as of 9 October 2026); not accepted or published.

ESC CONGRESS 2026 PRESENTATIONS

Munich, Germany · European Society of Cardiology

Zafar Ahmad (presenter), Nazia Perwaiz, Muhammad Salman Khan, Sidra Sultana, and Izharul Haq. "Synthetic ECG-guided temporal gating improves paediatric murmur grading, timing, and shape classification on the CirCor DigiScope dataset." 28 August 2026. Session: *Digital tools in practice: transforming clinical care*. ESC 365: esc365.escardio.org/presentation/326239.

Title as presented. Its results are superseded by the October 2026 patient-grouped evaluation, which found useful information in expert-reviewed phase alignment but did not support a benefit from the tested synthetic-ECG timing route. See the current research finding: zafar-ahmad-5359.github.io/research.html#current-evaluation.

Hamza Anjum, Nazia Perwaiz, Sidra Sultana, Tahira Anwar Lashari, and **Zafar Ahmad (presenter)**. "An explainable calibrated concept bottleneck model with evidence-constrained large language model reporting for multi-label valvular heart disease screening from 12-lead ECGs." 31 August 2026. Session: *Artificial intelligence-powered assessment of cardiac function*. Role: technical supervision and congress presentation. ESC 365: esc365.escardio.org/presentation/326198.

Manuscript statuses updated 9 October 2026. Congress titles reproduce the presented abstracts.